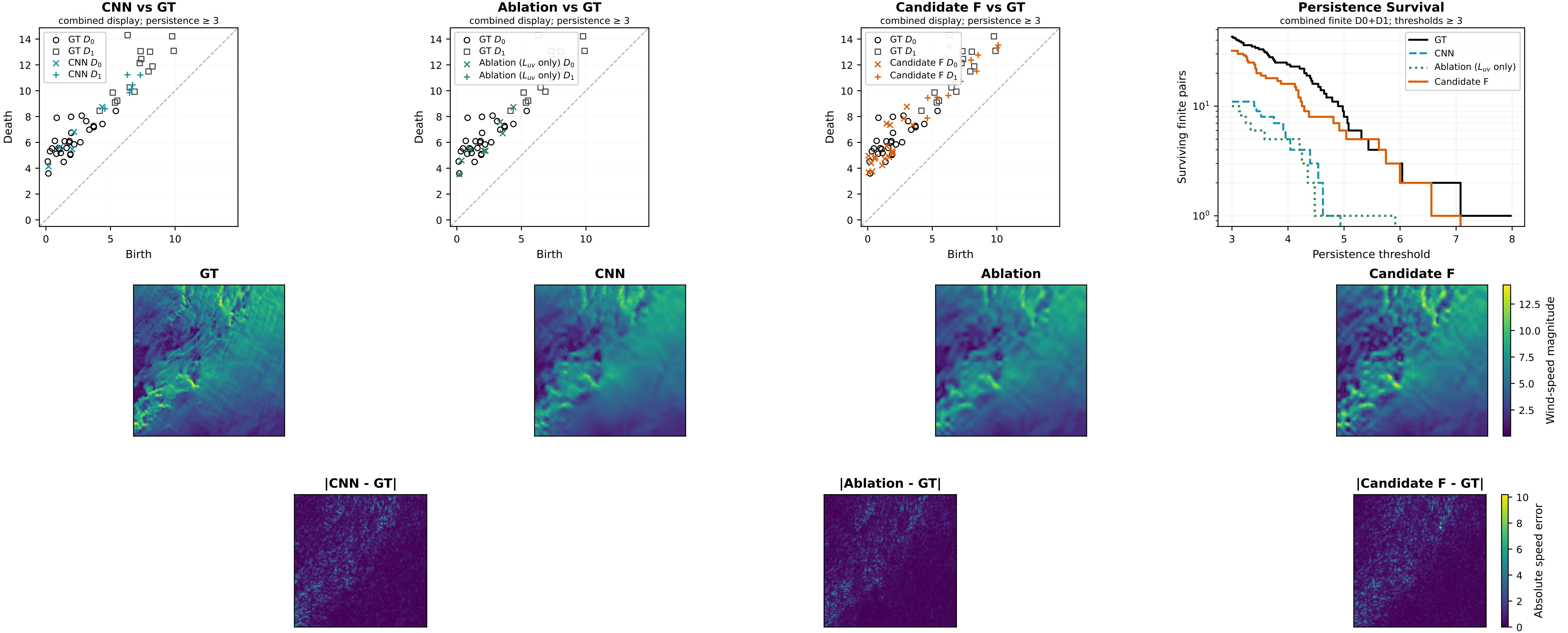
Sample 69: Topology-Aware Gains Beyond Matched Fine-Tuning

CNN = pretrained baseline; Ablation = matched L_{uv}-only fine-tuning; Candidate F = $L_{uv} + L_{grad} +$ repaired E2 supervision.



Topology chain (lower is better)

d_B: CNN 3.067 → Abl. 2.436 → F 1.233

$W_{2,\infty}$: CNN 14.572 → Abl. 15.256 → F 9.987

$W_{2,2}$: CNN 18.722 → Abl. 19.913 → F 12.323

Conventional fidelity gaps

F vs CNN (original closeness rank 27/168): |ΔPSNR|=0.775 dB, MAE=9.5%, RMSE=4.7%

F vs L_{uv} ablation: |ΔPSNR|=1.277 dB, MAE=17.2%, RMSE=16.3%

Matched-control interpretation

Candidate F is lower than the matched L_{uv}-only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.

Actual benchmark artifacts only. PD panels are display-thresholded at persistence ≥ 3; audited distances use all strictly positive finite pairs and preserve homology dimension. Field/error maps use the same 160×160 topology domain.